

Lab 02 Results – Fitting a curve to the dam

Numerical Methods BES6-M · Prepared by Garcia · Reservoir stage log, 288 readings, Jul 21 to Jul 23, 2026

Model

$$h(t) = c + a1 / (1 + \exp(-k1 (t - t01))) + a2 / (1 + \exp(-k2 (t - t02)))$$

The stage rises in one rapid pulse of about 7 m and then only partly recedes to a lower, flatter stage rather than settling at its peak, a shape a single logistic or Gompertz cannot reproduce because both are monotonic. A second, negative-amplitude logistic stacked on the first reproduces exactly that: one sigmoid for the inflow surge and an opposing sigmoid for the drawdown that follows it, each easing into a flat segment.

Fitted parameters (Levenberg–Marquardt)

Parameter	Value	Std. error	t statistic	p (two-tailed)
c (baseline, m)	14.3493	0.0097	1483.17	< 0.001
a1 (rise amplitude, m)	8.9158	0.7532	11.84	< 0.001
k1 (rise rate, 1/hr)	0.5117	0.0213	24.05	< 0.001
t01 (rise inflection, hr)	31.5198	0.0906	348.02	< 0.001
a2 (recession amplitude, m)	-3.7210	0.7594	-4.90	< 0.001
k2 (recession rate, 1/hr)	0.2831	0.0138	20.58	< 0.001
t02 (recession inflection, hr)	37.5465	1.0576	35.50	< 0.001

Goodness of fit

SSE	SST	R ²	s (m)	n	p	dof
1.8017 m ²	2034.08 m ²	0.99911	0.0801	288	7	281

Reading the residuals

The residual mean is 0.00000 m (essentially zero, no overall drift), but the scatter is not free of structure: sign changes only 17 times across 288 points -- far fewer than the roughly 144 runs pure noise would produce -- so the errors come in short streaks rather than flipping randomly, the signature of a shape mismatch rather than logger noise. The spread nearly triples during the surge window (std 0.1136 m) compared with the quiet pre-event segment (std 0.0908 m) and the settled post-event segment (std 0.0356 m), and the single largest residual is 0.2208 m at Jul-22 06:15 -- 22x the logger's own 1 cm resolution. The fit is good (R²=0.9991) but not perfect: the true rise is sharper than a smooth double logistic can trace, exactly where the model earns its worst marks.

Area under the fitted level

Quad integral	Quad abs. error	Trapezoid (raw)	Gap
1260.9708 m·hr	5.09e-07 m·hr	1260.9963 m·hr	-0.0254 m·hr (-0.0020%)

Quad integration of the fitted curve gives 1260.9708 m·hr (quad-reported absolute error 5.09e-07 m·hr). The trapezoid rule on the raw readings gives 1260.9963 m·hr, a gap of -0.0254 m·hr (-0.0020%) -- small because the fit tracks the data closely (R²=0.9991); the gap is exactly the area under the residual curve, positive where the fit overshoots the readings and negative where it undershoots. This number is a depth integrated over time in meter-hours, not a volume -- turning it into a volume would need the reservoir's stage-area curve, which this log does not contain.

1st/2nd derivative highlights

Maximum dh/dt = 0.960 m/hr at Jul-22 05:15 (t = 29.25 h from the first reading).

The second derivative swings positive during the pre-dawn surge as inflow accelerates, crosses zero at Jul-22 05:15 almost exactly where dh/dt peaks, then runs negative through the crest and into the recession -- meaning the reservoir was not just rising and falling, it was speeding up into the surge and slowing down out of it, and that acceleration information is invisible in the raw depth trace or the 1st derivative alone.